Predictive Maintenance in Electrical Machines: An Edge Computing Approach

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***Abstract*—Predictive maintenance is one of the premises of the Industry 4.0. A condition monitoring and fault diagnostic never seen before are possible with the new technologies embracing the digitalisation. This paper presents an edge computing approach for the predictive maintenance of electrical machines based on motor current signature analysis (MCSA), using Fast Fourier Transform (FFT) and a Support Vector Machine (SVM) classifier. A broken rotor bars and eccentricity fault detection are implemented on Arduino platform, providing a low-cost solution affordable for small and medium enterprises. The system has been tested in a lathe in one of the laboratories of the Mining and Industrial School of Almaden (UCLM) with satisfactory results.**

Keywords—Condition Monitoring, Predictive Maintenance, Fault Diagnosis, induction motors, edge computing

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I. INTRODUCTION

Being immersed in the fourth industrial revolution, Digitalisation, the use of new technologies and lines of research are emerging: Internet of Things, Big Data, everything Smart, Cloud Computing... all of them with the intention of digitalising information.

The origin of this digitalisation lies in Industry 4.0 (Industrie 4.0) [1], German Federal Government project to promote digitalization in manufacturing or networked, digitized production [2], where machines and products are seen as intelligent, networked components, which can exchange data locally, globally and across company borders.

The *Fourth Industrial Revolution*, or Industry 4.0, uses digital technologies (IoT, AI, sensors) to blend the physical and virtual [3].

The Industry 4.0 approach aims to [4]:

- Efficiently control distributed control systems through a partnership of autonomous units.
- Blending the physical world with the virtual world; where each physical element (sensors, products, raw materials, handling and transport machinery, operators) is represented by virtual units. This requires connectivity between each virtual element with the rest of the virtual and physical elements, as well as a control implementation from the virtual to the physical world.
- Decision-making, optimization and efficiency (cost-effectiveness, high performance and energy efficiency).
- Creation of new business models and service-based approaches to value creation.

The way to implement Industry 4.0, or the digitalization revolution, is through Cyber-Physical Systems (CPS), or as they are known in their more industrial version: Industrial Cyber-Physical Systems (ICPS) [5] or Cyber-Physical Production Systems (CPPS) [6], combined with the widespread use of the Internet of Things, (IoT) [7]. ICPS provide the necessary digital representation, smart service development and interoperability interfaces required to support the operation of flexible, network-based manufacturing environments, where embedded smart devices work together, via the IoT, and centralised control is transferred to a distributed smart network [8].

This digitalisation is having a major impact in other areas such as smart cities/buildings, smart energy systems, pollution monitoring, health tracking or intelligent transport systems. However, unfortunately, its adoption by industry has yet to be fully realised.

This paper focuses on one of the premises of Industry 4.0: predictive maintenance, and more specifically on predictive maintenance in electrical machines. According to the International Energy Agency report [9], around 50% of all energy generated in developed countries is consumed by electric induction motors. So, the interest in fault prediction and detection on these machines has increased significantly [10-12]. This paper presents an ICPS solution for the predictive maintenance of an electrical machine, aiming to reduce maintenance costs and equipment downtime by detecting equipment faults.

The remaining paper is organized as follows: Section II presents the commonly used techniques to perform a predictive maintenance in inductive electric motors. Sections III presents the parts of the ICPS system performing the predictive maintenance, paying special attention to provide an edge computing solution, i.e. a solution capable of computing data locally. Section IV shows the experimental results carried out to test the proposal. Finally, Section VI rounds up the paper with the conclusions and points out some future work.

II. STATE OF THE ART

The three-phase induction motor with squirrel cage rotor has become widespread in industry. Its use, as with all other types of motors, is exposed to various hostile environments, incorrect applications, and manufacturing defects. Internal defects such as short circuits, earth faults, broken rods or bearing failures are unavoidable. The same applies to defects caused by external agents or effects. In addition, operation in aggressive environments can accelerate ageing and make it more susceptible to the appearance of new faults, and the gradual deterioration caused by ageing itself will result in the potential for motor failure if these internal and external defects are not detected and corrected early enough.

Induction motors are protected by devices such as relays, circuit breakers, etc., but the performance of these protections is limited either to protecting the motor against

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an external problem or to a significant level of motor deterioration, when the fault is often irreversible. On the other hand, the demand from users to increase the reliability of electric motors is constantly growing, due to the importance of many applications, advances in technology and the need to obtain better operating results. In this situation, the possibility of knowing when a failure is starting to develop becomes essential in those applications which, due to their complexity, critical operation, cost of shutdown or repair, safety, do not allow the motor to fail unexpectedly.

Several diagnostic methods have been developed for three-phase induction motors. These methods are generally based on the analysis of the magnitudes available at the motor terminals or of those faults that can be observed by means of suitable devices, proceeding to their measurement under normal operating conditions of the motor or with the machine out of service.

A. *Out-of-service Diagnostics.*

These techniques are characterised by the fact that they are carried out when the machine is not in operation, in many cases requiring the motor to be switched off and even partially or totally dismantled. They are usually carried out during scheduled overhaul periods. Most of these methods are dedicated to observations on the degree of deterioration of the insulation system and on the state of the magnetic core. These methods include [13]:

- Measurement of insulation to ground: Insulation resistance, polarisation index, absorption current, tangent delta capacitance, loss factor, overvoltage tests.
- Insulation between turns: Shock wave test, insulation resistance, reverse sequence impedance, electrical circuit analysis, parameter estimation.
- Magnetic core analysis: with application of nominal or reduced flux.
- Bearing lubricant analysis: Physical characteristics, particle content.

B. *In-service Diagnostics.*

In-service diagnostic methods provide information on the state of the machine without the need for interruptions in operation. This is an important advantage, as each of them usually has a series of analysis utilities, normally in the form of software, which allows them to be used in predictive maintenance programmes, making their application extremely attractive.

The most important methods for diagnosing faults with Condition Monitoring (CM) are [10]:

- Stator current analysis or motor current signature analysis (MCSA) [14, 15].
- Vibration analysis [16, 17].
- Temperature measurement (thermography) [18, 19].
- Partial discharges.
- Gas analysis.
- Axial dispersion flux analysis
- Torque and speed fluctuations.
- Input power analysis.

The first three being the most important. In this paper MCSA is used as it is non-invasive, lower-cost and reliable technology to develop a condition monitoring and fault diagnosis in an induction motor.

C. *MCSA in induction machines*

The most studied faults in induction motors are bearing faults, stator faults, broken rotor bars, and eccentricity faults. Each of these faults have a characteristic frequency in the MCSA analysis of the induction motor. In this papers eccentricity and faults and broken rotor bars are covered.

1) *Eccentricity Faults*

Two different eccentricity faults can appear: the static and the dynamic. Fig. 1 shows both.

Static eccentricity is the distortion of the air gap size in which its minimum value remains in a fixed position. It occurs basically in two ways: either due to the oval shape of the stator or due to incorrect positioning of the rotor.

In the case of dynamic eccentricity, the point of minimum air gap rotates together with the rotor and thus changes its position over time. This type of eccentricity can be caused by an oval shape of the rotor or by the centre of rotation of the rotor being out of phase with the geometrical centre of rotation.

When a mixed eccentricity fault happens, sidebands of the fundamental frequency of the power supply appears. These sideband frequencies are given by [20]:

$$f_{ecc} = f_s \left[1 \pm m\left(\frac{1-s}{p}\right)\right] \tag{1}$$

Where f_s is the supply frequency, s is the slip, p is the number of pole pairs, and m =1,2,3,… This scheme compared to others does not need to know the mechanical characteristics of the motor.

2) *Broken Rotor Bars*

Broken rotor bars are one of the main ways in which a motor can fail, especially in high-power motors that frequently start under load. The starting transient is the one that causes the most stresses in the rotor bars, as during this transient the highest currents appear in these bars due to the difference in speeds between the rotor and the stator field.

When there are broken rotor bars in a motor, the rotor symmetry is destroyed which results in the appearance of a rotating field that generates harmonics in the stator current according to by [20]:

$$f_{brb} = f_s \left[k\left(\frac{1-s}{p}\right) \pm s\right] \tag{2}$$

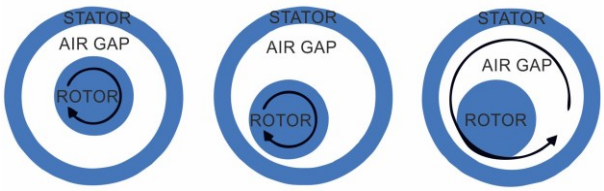


Fig. 1. Healthy motor (left), static ecentricity fault (centrer), and dynamic eccentricity (rigth)

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Where, due to the normal winding configuration, $k = p, 5p, 7p, 11p, 13p, \ldots$

As the influence of the main frequency is predominant, equation (2) can be simplified as:

$$f_{brb2} = f_s[1 - 2s] \quad (3)$$

III. Predictive Maintenance in the ICPS Era

A Cyber Physical Systems (CPS) integrates computational applications with physical devices, being designed as a network of interacting cyber and physical elements [4]. In contrast to embedded systems, where the focus is on computational elements hosted in stand-alone devices, CPS are mostly designed to be a larger network of interacting computational and physical elements [21]. When CPS are used in industrial environments emergence the ICPS.

Under the topic covered in this paper, the ICPS will comprised many components, but the related to the predictive maintenance can be summarized as the following two:

- Physical part. It includes all the sensors involved in the data acquisition necessary to carry out the predictive maintenance.
- Cyber part. It is the intelligent one, where the data coming from the physical world is analysed and interpreted. Here is where emerging technologies such as Big Data and Artificial Intelligent are applied.

At this point two different approaches can be taken: Edge or Fog/Cloud Computing

1) Fog/Cloud computing.

These are scenarios where the data is collected locally and send it to the fog or the cloud for further processing. This setup provides high computing power. Some researchers are focused on test or develop new machine learning models based on simulation environments such as Matlab/Simulink [22], or works where datasets are modified including possible failures to test the performance of different identification techniques [23]. It is common that machines are located at fixed position, although there are exceptions. In [24] is presented the predictive maintenance of the mobile machinery present in a mining complex through Bluetooth Low Energy vibration sensors.

2) Edge computing.

When the cyber and the physical part are located close to the electrical machine to be tested. This approach is difficult to implement in industrial environments due to the computing effort required to process the data, leading to the necessity of have a personal computer nearby.

In controlled environments, as laboratories, this setup is very common: data acquisition cards and equipment to process the information, mainly a personal computer with specialized software, where it is possible to provoke faults in the electrical machine and study the consequences. In [15] is used a NI DAQ to acquire the information from a current sensor an test different pre-filter technics to prepare the data. In [25] a NI DAQ, a current probe and a tachometer are used to test an induction motor and test different machine learning methods, neural network and multimodal deep learning. Ref. [14] proposed a deep network-based to study the features of thermograms, with the help of a thermal camera. These setups are permissible in an academic environment to perform tests and analyse the results to provide new techniques, but in and industrial environment can be difficult to implement.

There are few examples where the edge computing is taken to its maximum expression. In [26] is proposed a bearing fault detection through vibration analysis by means of ultra-low power wireless sensors. The sensors acquire and analyse the data and only send information regarding the good/bad state of the machine to the rest of the ICPS. This application is possible thanks to the use of single axis accelerometers and the development of a new light weight neural network classifier.

An edge computing solution for predictive maintenance based on and ICPS architecture is presented in the following sections.

B. ICPS Physical Part.

It is necessary to count with a data acquisition system in order to carry out a predictive maintenance in electrical machines focused on the detection of airgap eccentricities and broken rotor bars by means of MCSA techniques.

As the faults that are going to be analysed appears at the sidebands of the main frequency, 50 Hz, it is chosen a sampling rate of ten times the main frequency, 500 Hz. This sampling frequency broadly meets the Shannon-Nyquist frequency, so problems of aliasing or faulty reconstruction signals are avoided.

An Arduino Uno board has been chosen for the data acquisition. It has 6 analog input pins. This analog-to-digital converter (ADC) has a default sampling rate of approximately 9600 Hz. This sampling rate is due to the 16MHz of the ADC clock of the Arduino's microcontroller, ATMega328P, and the prescale factor set to 128, as default. If the Arduino's environment functions are avoided, it could be possible to achieve a sampling rate of 615 kHz, decreasing the prescale factor to 2 and using interrupts. For the application at hand is enough the default sample rate.

Once is defined the sampling rate, and checked the chosen ADC provides it, it is time to accommodate the signal of the power grid (230 VAC line to ground voltage) feeding the induction motor to the analog input (0-5 VCC).

To measure the current consumed by the induction motors it is chosen a non-invasive clamp sensor, model SCT-013-030 by YHDC®, installed as Fig.2 shows. SCT-013-030 is a current transformer with a range of 0-30 A providing an output voltage from 0 to 1V. As alternate current has positive and negative values it is necessary to adapt the range to the positive range of Arduino. A 0,5 voltages offset has been added using a voltage divider. With the aim of increasing the resolution in the data acquisition, a single-supply instrumentation amplifier configured with a gain of 4 is included, specifically the AD623 by Analog Devices® and, after that, an OP113 operational amplifier configured in voltage follower, as AD623's manufacturer recommends. Therefore, the final characteristic curve of the sensor is given by:

$$i(t) = \frac{15}{3.5}(v(t) - 4) \quad (4)$$

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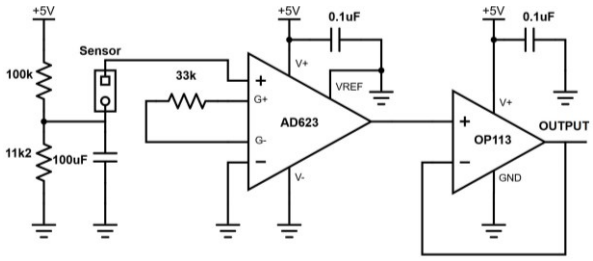


Fig 2. Signal conditioning circuit

As the ADC of Arduino has 10 bits of resolution, it is possible to reach a resolution in the measurement of 0,036 A. Taking this information into account the characteristics curve for the sensor attending to the readings of the ADC will be:

$$i(t) = \frac{15}{3.5}\left(\frac{5}{1023}ADC(t) - 4\right) \quad (5)$$

C. ICPS Cyber Part.

For the cyber part of the edge computing is proposed to continue using the Arduino Uno. The cyber part is in charge of process the data coming from the sensors, extract the specific features, classified them, and share the status of the machine to the rest of the ICPS, as Fig. 3 shows.

1) Signal Processing.

Raw data coming from the sensors is a time-based signal. A frequency-based signal is needed to extract the convenient information. For that, it is possible to use Fast Fourier Transform (FFT) [16, 27], wavelet [14] or Hilbert–Huang Transform [28].

FFT is the chosen method in this paper. So, the signal has to be windowed:

$$i_R^*(t) = \begin{Bmatrix} i_R^*(t) & for\ t\ \epsilon\ [0,T] \\ 0 & Otherwise \end{Bmatrix} \quad (6)$$

And then compute the Fourier Transform:

$$I_R^*(\omega) = \int_{-\infty}^{\infty} i_R^*(t)\, e^{-j\omega t} dt \quad (7)$$

The ArduinoFFT library is used to compute the FFT of the acquired signal.

2) Feature Selection.

As stated in previous section II.C, it is necessary to measure the sidebands of the main frequency to detect eccentricity or broken rotor bars faults. So, the amplitude of that frequencies will be the features, ϕ, chosen for the classification of the motors as healthy or faulty in the map Φ. Therefore, each feature is computed as:

$$\phi^* = \Phi\left(i_R^*(t)\right) = |I_R^*(\omega)| \quad (8)$$

3) Classification.

The prediction methods use the features to classify the motor as healthy or faulty. Prediction methods can be classified as data-driven methods and model-based methods [29]. Data-driven methods attempt to derive the degradation processes of machinery from measurements using machine learning techniques. It is common to find solutions based on Artificial Neural Networks (ANN), Fuzzy Logic and Neuro-Fuzzy Systems (NFSs), Evolutionary Algorithms (EAs), Support Vector Machine (SVM) [30], among others.

Model-based methods rely on establishing empirical or physical models to describe the degradation processes of the machine and update the model parameters according to real-time information from sensors.

The normal procedure in SVM is first to train the classification model with a tagged dataset including the involved classes to, then, test the model with another dataset, in this case with untagged data, and analyse the performance.

Intuitively, an SVM is a model that represents sample points in a space, sorting them into classes that characterizes 2 subspaces separated by a hyperplane, as is shown in Fig 4. The hyperplane is defined as the vector between the 2 points of the 2 classes, closest to the called support vectors. The support vectors define the boundaries of the subspaces. The distance between support vectors of each subspace is the margin. As there are many hyperplanes able to classify the samples, it is common to choose the one with the largest margin.

New arriving samples are transformed into the classification space and, depending on their location, they will belong to a subspace and, therefore, can be classified into one or the other class. For the case in hand as “faulty” or “healthy”.

Although it is possible to separate non-linear regions using kernel functions, the chosen hyperplane, $H(w,b)$, will follow the common expression:

$$H(w,b) = w \bullet \phi - b = 0 \quad (9)$$

Where ω is the weight vector, ϕ is the feature in the featured map Φ, and b is the bias term, a point in the space.

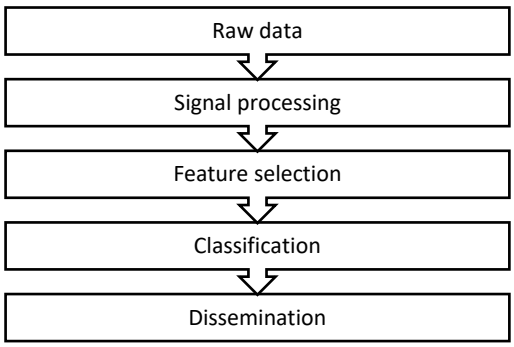


Fig 3. Flowchart of the cyber part

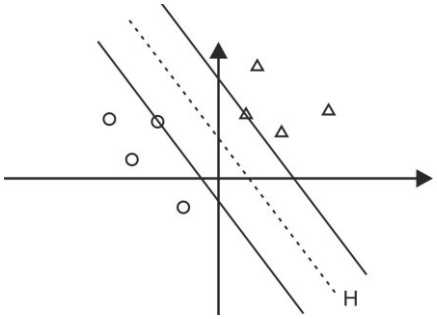


Fig 4. Hperplane of a SVM classifier

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In this paper it is used the SVM adaptation proposed in [27] for the case of broken rotor bar faults. Instead of train the model, it is defined theoretically for a healthy and a faulty motor, so, the training phase is theoretically solved, leading to the following solution:

$$w = \frac{2(\phi^h - \phi^f)}{||\phi^h - \phi^f||^2} \quad (10)$$

$$b = \frac{(\phi^h - \phi^f)(\phi^h + \phi^f)}{||\phi^h - \phi^f||^2} \quad (11)$$

Where ϕ^h denotes the feature value of a healthy motor, and ϕ^f the value for a faulty one. Both features have to be computed theoretically according to the characteristics of the electrical motor.

4) Dissemination.

Once the system has estimated the status of the machine, this information will be share with the rest of the ICPS controlling the system. Depending on the scenario will be useful to use LAN or Wireless communications to disseminate the information. One feasible solution having and Arduino Uno is to include a communication shield, but, in order to do not interfere in the normal operation of the microcontroller would be better to use a UART-LAN/WIFI interface module that executes all the communication protocol, and let Arduino only to send the information through the USART port.

IV. Experimental Results

The proposed fault detection method based on Arduino Boards has been tested on one of the lathes of the Mechanical Engineering Laboratory at the Mining and Industrial School of Almadén (UCLM-Spain), specifically a Pinacho A.75 (Fig. 5) equipped with a three phases electrical motor AEG AM 100 LZ 4 with the characteristics shown in Table I.

According to the information presented in Table I, it is possible to calculate the slip, s, by:

$$n_s = \frac{60 \cdot f}{p} = \frac{60 \cdot 50}{2} = 1500rpm \quad (12)$$

$$s = \frac{n_s - n}{n_s} = \frac{(1500 - 1420)\,rpm}{1500\,rpm} = 0.053 \quad (13)$$

With this information and using equations (1) and (3) it is possible to determine the sidebands around the main frequency that will advise of:

- a broken bar (3): $44.667\ Hz$.
- an eccentricity fault (1): $26.333\ Hz$, $73.667\ Hz$, $121\ Hz$, and $168{,}333Hz$.

First, a commercial power analyser, CIR-e3 by Circutor®, was connected in the power supply of the machine to perform a first check of the amplitude of the featured frequencies. Unfortunately, this device only provides information related to the common harmonics, neither sidebands nor graphical results are presented to have an idea of the state of the machine. The voltage wave presents harmonics of 3rd, 5th and 11th order, related to frequencies of 150 Hz, 250 Hz and 550 Hz, respectively. So, apart from the subjective opinion of the operator of the lathe, guessing about the perfect condition of it, there is no more extra information.



Fig 5. Pinacho A.75 tested

When using ArduinoFFT library it is necessary to define the sampling frequency and the number of samples. It is necessary to reach a good frequency resolution, that states how far are two consecutive frequencies in the spectrum. The frequency resolution is calculated by:

$$Res_f = \frac{f_{sampling}}{samples} \quad (14)$$

There are two options to improve the resolution: use a smaller $f_{sampling}$ or increase the number of samples. The first one is restricted by the Shannon-Nyquist frequency. The second one is restricted by the resources of Arduino board. So, for a 500 Hz $f_{sampling}$ and 128 samples (restricted by the 2K of RAM in Arduino Uno) the frequency resolution is about 3.9 Hz/division; 3.9 Hz from bin to bin. This value is worse than the required, as it does not allow us to collect enough useful information between slots. For the case of an Arduino Mega (8K of RAM) is possible to reach 512 samples and a frequency resolution of 0.97 Hz/division. A second configuration (120 Hz and 512 samples) for Arduino Mega can improve the frequency resolution to 0.23 Hz/division. These last configurations have been used in the test.

The amplitude of the featured frequencies are defined as the chosen features, ϕ, for the classifier to estimate if there is a broken bar in the motor or an eccentricity fault.

The hyperplane of the classifier has been defined from $\phi^h = 0.2$ and $\phi^f = 0.4$ as $H(w, b) = -10 \bullet \phi + 3 = 0$ leading to classifying ϕ as healthy when $H > 0$ and faulty when $H < 0$.

After testing with the two-hardware configuration for the data collection, and locally computing the FFT in Arduino, the spectrums shown in Fig. 6 where achieved. Taking each of the features and using the SVM classifier leads to affirm that the motor of the lathe is healthy as there are no pikes at the featured frequencies. This information can be then share with the rest of the components of the system.

TABLE I. Characteristics of the Electrical Motor Tested

Description	Value
Power	4 kW
Frequency	50 Hz
Voltage	220 / 380 V
Current	11.8 / 6.8 A
Number of revolutions	1420 rpm
Number of poles	4

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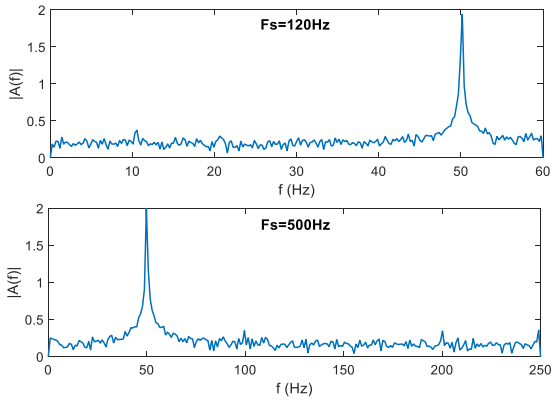


Fig 6. FFT results with Fs=120 Hz (up) and Fs=500 Hz (down)

V. CONCLUSIONS

Predictive maintenance, under the Industry 4.0 umbrella, brings many benefits for companies. In this paper a predictive maintenance system over edge computing is proposed. This predictive system has been conceived as an ICPS being part of another ICPS bigger. The physical and the cyber part has been defined and build over the Arduino platform, specifically the Arduino Mega board.

The system develops well known MCSA maintenance techniques for electrical motors in the edge, without the necessity of further analysis in the cloud. Two failures detections have been deployed: eccentricities and broken rotor bars.

The proposed solution is affordable for small and medium enterprises, allowing them to become part of the new Digitalisation paradigm and take advance of its benefits.

Future work is related to include more complex classifiers, based on information coming from different sensors and not only detect the fault, but to work on prognosis.

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